

DR. THALES A. GUTCKE
Postdoctoral fellow

University of Hawai'i
Institute for Astronomy
640 N A'ohōkū Place, Hilo, HI, 96720
gutcke@hawaii.edu
thalesada.github.io
ORCID: [0000-0001-6179-7701](https://orcid.org/0000-0001-6179-7701)

Born on March 11, 1988 in Bodrum, Turkey
Citizenship: Germany & USA

I'm a theoretical astrophysicist and former Hubble Fellow at the University of Hawai'i's Institute for Astronomy. My research centers on galaxy formation, particularly small, ancient dwarf galaxies. I use these systems to explore fundamental processes such as star formation and feedback that shape galaxies, but also to constrain Population III stars, the origin of globular clusters and alternative dark matter models. I initiated and lead the development of LYRA, a ultra high-resolution cosmological simulation that resolves every massive star and supernova from first principles. Using my large computing time grant of 40 million CPU hours, I have employed LYRA to create a suite of Local Group dwarf analogs that match various scaling relations. Before coming to Hawai'i, I held fellowships at Princeton University and the Max Planck Institute for Astrophysics. I actively collaborate with major observational programs using JWST, Gemini, and the VLT, and I serve on review panels for NASA and other agencies. Beyond research, I enjoy mentoring students, engaging in public outreach, and bridging theory and observation to better understand our cosmic origins.

ACADEMIC POSITIONS

11/2024 - 12/2027	Postdoctoral fellow, Institute for Astronomy, University of Hawai'i (part-time)
10/2024 - 10/2026	Maternity leave (including part-time)
11/2022 - 11/2024	Hubble fellow, Institute for Astronomy, University of Hawai'i
11/2021 - 11/2022	Hubble fellow, Princeton University
11/2018 - 11/2021	Independent fellow, Max Planck Institute for Astrophysics, Garching
05/2018 - 11/2018	Postdoctoral researcher, Max Planck Institute for Astronomy, Heidelberg
01/2016 - 05/2018	Visiting scholar, New York University Abu Dhabi

EDUCATION

04/2014 - 04/2018	PhD in Astrophysics, Max Planck Institute for Astronomy, magna cum laude
09/2011 - 03/2014	MS in Physics, Max Planck Institute for Astronomy
09/2008 - 08/2011	BS in Physics, University of Heidelberg

REFERENCES & COOPERATION PARTNERS

Prof. Volker Springel (postdoc adviser)	volker@mpa-garching.mpg.de
Prof. Andrea Macciò (PhD adviser)	maccio@nyu.edu
Prof. David Sanders (postdoc adviser)	sanders@hawaii.edu
Prof. Christoph Pfrommer (collaborator)	cpfrommer@aip.de
Prof. Azadeh Fattahi-Savadjani (collab.)	azadeh.fattahi@fysik.su.se
Prof. Carlos Frenk (reference)	c.s.frenk@durham.ac.uk

GRANTS & FELLOWSHIPS

2026-2027	JWST theory grant (\$140 000)
2021 - 2024	NASA Hubble Fellowship (~\$450 000)
2022	Princeton Scientific Writing Fellowship (includes 1 semester salary)
2021	Lyman Spitzer, Jr. Fellowship, Princeton (declined)
2021 - 2024	40 million core hours (equivalent to €1.6 million through AWS) at Leibniz Supercomputing Center (LRZ), Germany
2018 - 2021	20 million core hours (equivalent to €800 000 through AWS) at Max Planck Computing and Data Facility (MPCDF)
2018	MPA independent research fellowship (3 years salary)
2018	CCA/CfA joint postdoctoral fellowship (declined)

2017	Funding to host 12-week summer student (salary + expenses) from German Academic Exchange Service (DAAD)
2014	International Max Planck Research School (IMPRS) fellowship
2014	Full graduate scholarship (4 years salary) from German Research Council (DFG) Collaborative Research Center “The Milky Way” (SFB 881, A1)

COLLABORATIONS

2022 -present	Initiator, lead developer and PI of LYRA collaboration for dwarf galaxy simulations
2024	REQUIEM and EREBUS collaborations on high redshift quasars
2022 - present	Member of All Sky Automated Survey for SuperNovae (ASAS-SN)
2021 - present	PI of computing allocation (Leibniz Center/SuperMUC-NG, 40 M c/hrs)
2020 - 2021	Co-I of <i>Virgo</i> computing allocation (DiRAC/COSMA, 30 M c/hrs)
2014 - 2018	Member of NIHAO simulations collaboration

OBSERVING EXPERIENCE

2024	Co-I on JWST Cycle 3 program, Survey with 627.7 hours awarded
2023	Co-I on Gemini-N/GNIRS, 28 hours awarded
2023	Co-I on Gemini-N/GNIRS, 15 hours awarded
2023	Co-I on ESO-VLT/MUSE Large Program, 156 hours awarded
2023	Co-I on JWST Cycle 2 proposal, 13.9 hours awarded (includes €200k)
2022	PI on GEMINI-N proposal, 1.7 hours awarded
2017 - 2024	Co-I on 5 ESO-VLT/MUSE proposals (> 200 h awarded)
2017	16 nights observing at MPIA/ESO 2.2m, La Silla, Chile

TEACHING & MENTORING

2026	Advisor for undergraduate Elizabeth Olsen, U. Hawai‘i
2023 - 2025	Advisor for graduate student Finn Giddings, U. Hawai‘i
2023	REU supervisor for Keaton Donaghue, U. Hawai‘i
2023	Taught select lectures, <i>Computational Physics & Astronomy</i> , U. Hawai‘i at Hilo
2022	Taught graduate course, <i>Proposal Writing for Quantitative Disciplines</i> , Princeton
2022	Attended <i>Pedagogy in Practice</i> , teaching workshop, Princeton
2021 - 2022	Co-advisor of graduate student Jessica May Hislop, MPA
2020 - 2021	Co-advisor of graduate student Alejandra Fresco, MPA
2017	RISE summer research supervisor for Rokas Zemaitis, MPIA, Heidelberg
2015	M.S. physics lab instructor, University of Heidelberg
2014	B.S. physics lab instructor, University of Heidelberg

SERVICE ACTIVITIES & COMMISSIONS OF TRUST

2024	SOC member for <i>The First Gigayear(s)</i> conference, Hawaii
2023	NASA Astrophysics Theory Program review panel member
2023	NASA Hubble space telescope review panel member (Cycle 31)
2022	Volunteer at ‘Imiloa Astronomy Outreach Center, Hilo, Hawai‘i
2022	NASA Community College Network, <i>subject matter expert</i>
2021	<i>Future Leader</i> at AURA annual meeting
2020 - present	Reviewer for ANR and DiRAC computing proposals
2018 - present	Referee for MNRAS
2019 - 2021	Journal Club Coordinator, MPA, Garching
2017	Equal Opportunity Officer, MPIA, Heidelberg
2014 - 2016	Graduate student representative, MPIA, Heidelberg

INVITED TALKS

May 2026	General Seminar, Arcetri Observatory, Florence, Italy
Jul 2025	Conference speaker, European Southern Observatory, Germany
Jun 2025	Conference speaker, University of Heidelberg, Germany
Jun 2025	EREBUS COSMOS-3D collaboration meeting, University of Bologna, Italy
Sep 2024	Hubble Symposium, NASA/STScI, Caltech, USA
Jul 2024	Dwarf Galaxies, Star Clusters, and Streams in the LSST Era, U. Chicago, USA

Jun 2024	Colloquium, Scuola Normale Superiore, Pisa, Italy
Apr 2024	EPFL/Geneva Observatory Joint Colloquium, Switzerland
Apr 2024	Invited Seminar, DIPC, San Sebastián, Spain
Apr 2024	Invited Seminar, New York University, USA
Oct 2023	IfA Colloquium, Institute for Astronomy, University of Hawai‘i, Mānoa, USA
Sep 2023	Hubble Symposium, Harvard University, USA
Sep 2023	CGM@ND II, Kylemore Abbey, Ireland
Apr 2023	Königstuhl Colloquium, Max Planck Institute for Astronomy, Heidelberg
Mar 2023	Bahcall Lunch, Princeton University, USA
Nov 2022	Colloquium, Department of Astronomy, Ohio State University, USA
May 2022	Theory Seminar, Canadian Institute for Theoretical Astrophysics, Toronto
Apr 2022	TAC Seminar, Theoretical Astrophysics Center, UC Berkeley, USA
Mar 2022	Joint Astrophysical Colloquium, INAF - University of Bologna, Italy
Feb 2022	Joint Colloquium, Steward Observatory/NSF’s NOIRLab, Tucson, USA
Feb 2022	IfA Colloquium, Institute for Astronomy, University of Hawai‘i, Mānoa, USA
May 2021	Seminar, ICRAR/UWA, Australia
Mar 2021	Lorentz Center Workshop, Leiden, The Netherlands
2016 - present	<i>Frequent public speaker</i> , including Dark Matter Day, MPIA open house day and Science March

SCHOOLS & INTERNSHIPS

Aug 2020	Parallel Programming with MPI/OpenMP, ETH Zurich
Sept 2016	11th Heidelberg Summer School on Astrostatistics & Data Mining
Jan 2016	HGSFP Winterschool, Obergurgl, Austria
Sept 2015	Astro Hack Week, New York University
Aug 2013	Adaptive Optics Summer School, University of California, Santa Cruz
April 2012	Calibration and integration of the Yale Doppler Diagnostic Facility